Towards the “Model Building in the Humanities through Data-Driven Problem Solving” based around the Japanese Literary Studies

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Introduction

The National Institute of Japanese Literature (NIJL) is a fundamental research institute for Japanese literature that has amassed a large collection of Japanese classical books and makes them available for use by researchers in various fields, including Japanese literature. The term “classical books”(古典籍) refers to all books (written works) that were created in Japan before 1868. This paper introduces the large-scale DH research project that the NIJL will be carrying out over the next ten years.

Background to the Large-scale Research Project

From 2014 to 2023, the NIJL has been implementing the “Project to Build an International Collaborative Research Network for Premodern Japanese Texts” #NIJL-NW Project# project, and has been building the “Database of Pre-Modern Japanese Works', which makes digital images of Japanese classical texts available to the public. As a result of this research project, 300,000 classical texts have been digitized and made available for use. In March 2023, the “Database of Pre-Modern Japanese Works” was integrated with the “Union Catalogue of Early Japanese Books”, a union catalog of classical texts created by the NIJL, and was reborn as the “Union Catalogue Database of Japanese Texts”.

The successor to the research project mentioned above is the “Model Building in the Humanities through Data-Driven Problem Solving: Cultivating Next-Generation Humanities Research through Creation and Deployment of New Data Infrastructure” (abbreviated to the DDH Project, 国文学研究資料館 2021), which started in 2024 as a 10-year project. The aim of this project is to establish and promote a new type of humanities research that works with other academic fields through the research and utilization of data, based on the achievements of the Union Catalogue Database of Japanese Texts.

The Four Research Areas of the DDH Project

The DDH Project has established four research areas (福島 / 菊池 2023). These are “Construction of Data Infrastructure”, “Development of Humanities Data Analysis Technology”, “Material Analysis and Interpretation”, and “Expansion from Content Analysis”, and the aim is to utilize the classical book data accumulated in the Union Catalogue Database of Japanese Texts.

The research area of “Construction of Data Infrastructure” is to apply OCR processing to the image data in the Union Catalogue Database of Japanese Texts and to construct a large-scale text data set of Japanese classical books. In addition, in order to enhance the Union Catalogue Database of Japanese Texts to a data infrastructure, we aim to provide classical book data to a wide range of users.

In the research area of “Development of Data Analysis Technology for the Humanities”, we are particularly focusing on the TEI/XML data conversion of classical texts. In addition, we are providing a learning environment for analysis methods using XML data and developing a viewer. We are also working on the development of a text mining analysis environment for plain text data of classical texts and the development of search technology for non-text information such as illustrations in classical texts.

In the “Material Analysis and Interpretation” area, we aim to digitize and extract the material characteristics of classical books, and then analyze them. Specifically, this involves research into the analysis of classical books using equipment such as high-resolution digital microscopes and handheld X-ray fluorescence optical analyzers. Through this, we also aim to develop, accumulate and utilize new metadata for classical books.

The “Expansion from Content Analysis” aims to establish environmental humanities by utilizing data obtained from the three research areas mentioned above, including the study of disaster prevention for classical texts, which utilizes data

on classical texts for disaster prevention research, and the study of literary tourism resources, which contributes to regional development by utilizing classical texts as tourist resources. All of these aim to solve issues in human society, going beyond the study of Japanese literature.

Current Status and Future of the Project

The DDH Project has just started over 30 joint research projects in the four research areas mentioned above. As the NIJL continues to develop the DDH Project over the next 10 years, collaboration with DH researchers and GLAMs from various countries and on various themes will be of great significance.

Bibliography

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